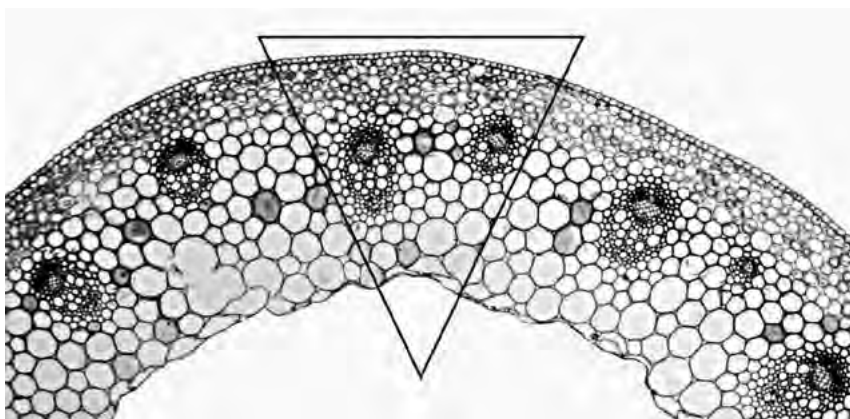


Answer ALL questions.

Write your answers in the spaces provided.

- 1** A student made observations of plant cells and tissues through a microscope.
- (a) (i) Photograph A shows part of a transverse section of a stem at a magnification of $\times 40$.



Source: John Addis

Photograph A

Draw a low-power plan of the area within the triangle shown on Photograph A **and** label two tissues on your drawing.

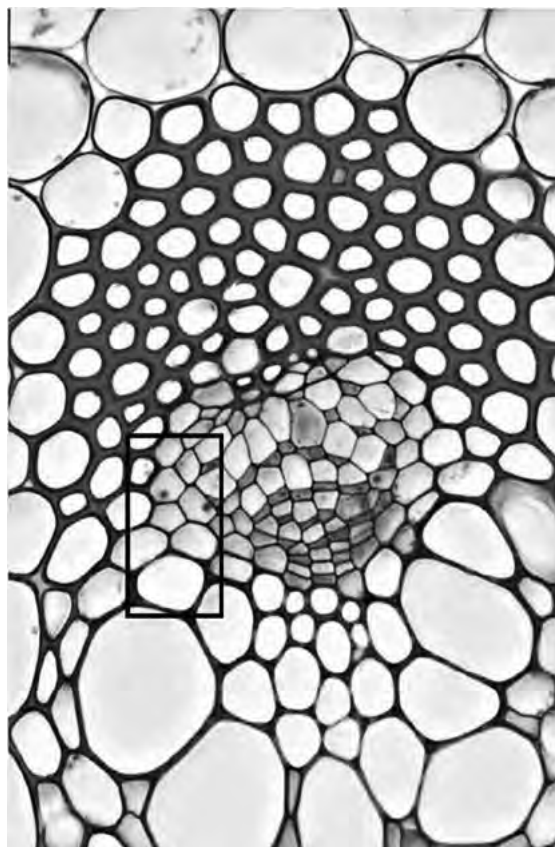
(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- (ii) Photograph B shows part of a transverse section of the same stem at a magnification of $\times 400$.



Source: John Addis

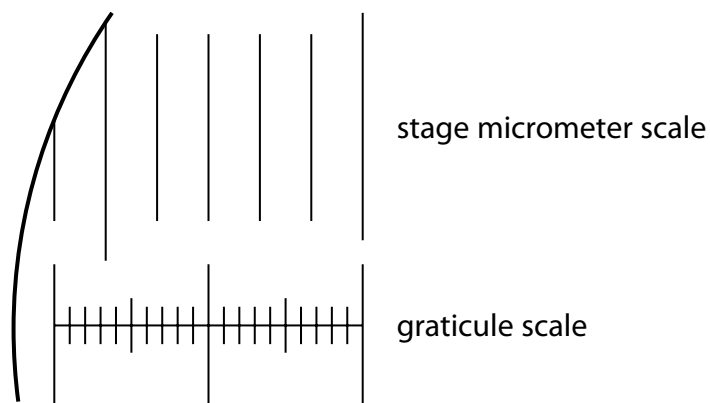
Photograph B

Draw the cells within the box shown on Photograph B.

(3)

(b) The diagram shows part of a stage micrometer scale and part of a graticule scale.

The smallest stage micrometer division is $10\text{ }\mu\text{m}$.



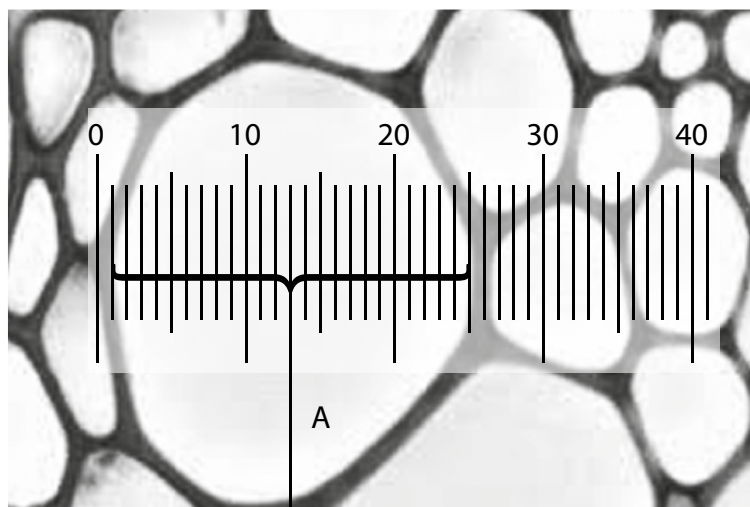
(i) Calculate the size of one small division of the graticule scale.

(2)

Answer..... μm

- (ii) Using your answer to (b)(i), determine the internal diameter of the part labelled 'A'.

(2)



graticule scale

Source: John Addis

internal diameter of part A

Answer..... μm

(Total for Question 1 = 10 marks)